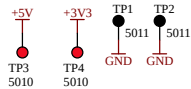
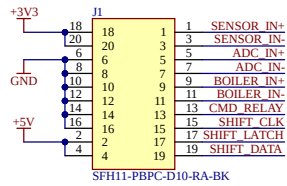
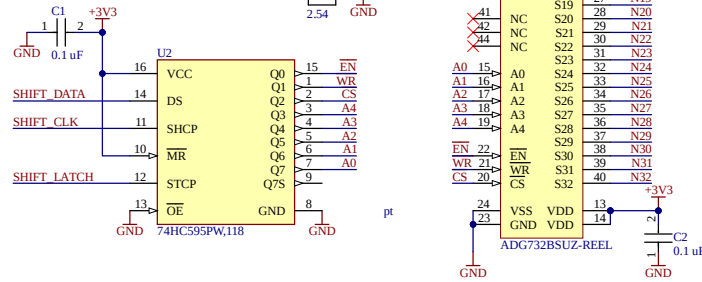


BOARD TO BOARD CONNECTOR

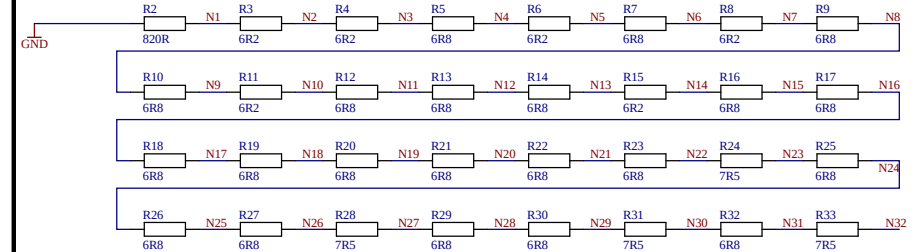


MUX CMD SIMULATION

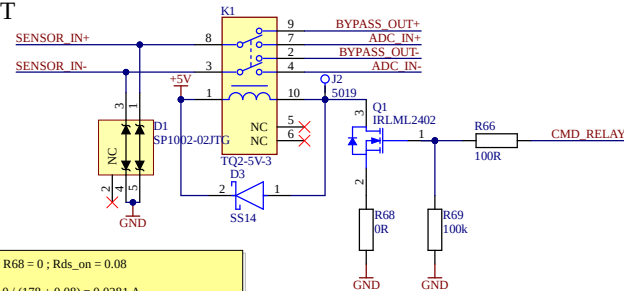
$$I_{dyn} = CPD \cdot VCC - I_i + \Sigma (CL \cdot VCC \cdot f_o) \\ = 115e-12 \cdot 3.3 - 1.0e6 + 40e-12 \cdot 3.3 \cdot (1.0e6/8) \\ = 0.3795e-3 + 0.0165e-3 \\ = 0.3960e-3[A] = 0.396[mA] \text{ par composant}$$
$$I_{total_max_par_composant} = I_{dyn} + ICC_max = \\ 0.396[mA] + 0.080[mA] = 0.476[mA]$$



RESISTOR NETWORK SMD



BYPASS INPUT

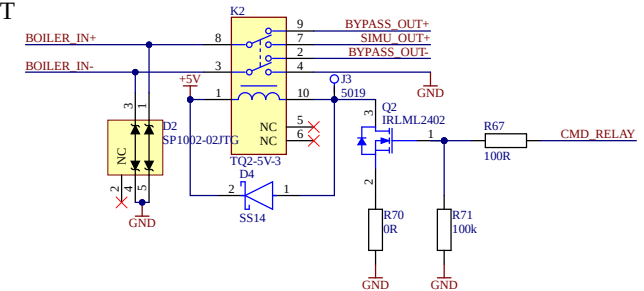


Data: $V = 5.0$; $R_{coil} = 178$; $R_{68} = 0$; $R_{ds_on} = 0.08$

$$I = V / (R_{coil} + R_{ds_on}) = 5.0 / (178 + 0.08) = 0.0281 \text{ A}$$
$$P_{total_5V} = V \cdot I = 5.0 \cdot 0.0281 = 0.1405 \text{ W}$$
$$P_{coil} = I^2 \cdot R_{coil} = 0.0281^2 \cdot 178 = 0.1404 \text{ W}$$
$$P_{MOSFET} = I^2 \cdot R_{ds_on} = 0.0281^2 \cdot 0.08 = 0.000063 \text{ W}$$

OFF state: $I \approx 0 \text{ A}$; $P \approx 0 \text{ W}$

BYPASS OUTPUT



ObloSolution * * * *	Mod.	a		Dessiné	LMS	19.08.2025	A3	 	Echelle :
		b		Contrôlé	*				_____
		c							
		d		Remplace					
Nom du projet		RegulationChauffage					No	2510B	

Line #	Name	Description	Designator	Revision ID	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	0.1 uF	CAP CER 0.1 uF 10V X5R 0805	C1, C2			2	Yageo Group	CC0805KRX7R9BB104	Volume Production	Farnell	3019949	0.03003	0.30028
	SP1002-02JTG	16V Pince 2 A (8/20µs) Ipp TVS - Diodes Montage en surface SC-70- 5	D1, D2			2	Littelfuse	SP1002-02JTG	Volume Production	DigiKey	F4238CT-ND	0.37	0.74
	SS14	DIODE SCHOTTKY 40V 1A SMA	D3, D4			2	onsemi	SS14	Volume Production	Farnell	1467537	0.26592	1.33
	SFH11-PBPC-D10-RA-BK	No Description Available	J1			1	Sullins	SFH11-PBPC-D10-RA-BK	Volume Production	Digikey	S9205-ND	1.16	1.16
	5019		J2, J3			2	Keystone	5019	Volume Production	Digikey	36-5019CT-ND	0.37	0.74
	TQ2-5V-3	RELAY TELECOM DPDT 1A 5VDC	K1, K2	CMP-05607- 000142-1	Up to date	2	Panasonic	TQ2-5V-3	Volume Production	Mouser	769-TQ2-5V-3	3.33	6.66
	IRLML2402		Q1, Q2			2	UMWIRLML2402TR	UMWIRLML2402TR	Manual Solution	Digikey	4518-UMWIRLML2402TR- ND	0.43364	0.86727
	200R	200 Ohms 0.5W, 1/2W PC Pins Through Hole Trimmer Potentiometer Cermet 1.0 Turn Side Adjustment	R1			1	Bourns	3386H-1-201LF	Volume Production	Farnell	9354930	1.78	1.78
	820R	820 Ohms ±1% 0,125W, 1/8W Pavé résistif 0805 (2012 métrique) Couche épaisse - RC0805FR- 13820RL	R2			1	Yageo Group	RC0805FR-13820RL	Volume Production	DigiKey	13-RC0805FR-13820RLCT- ND	0.1	0.1
	6R2	RES 6.2 OHM 1% 1/8W 0805 RC0805FR-076R2L	R3, R4, R6, R8, R11, R15			6	Yageo Group	RC0805FR-076R2L	Volume Production	Farnell	3952295	0.03663	0.36632
	6R8	RES 6.8 OHM 1% 1/8W 0805	R5, R7, R9, R10, R12, R13, R14, R16, R17, R18, R19, R20, R21, R22, R23, R25, R26, R27, R29, R30, R32			21	Yageo Group	RC0805FR-076R8L	Volume Production	DigiKey	311-6.80CRCT-ND	0.019	0.399
	7R5	RES SMD 7.5 OHM 1% 1/8W 0805 AC0805FR-077R5L	R24, R28, R31, R33			4	Yageo Group	AC0805FR-077R5L	Volume Production	DigiKey	13-AC0805FR-077R5LCT- ND	0.027	0.27
	100R	RES 100ohm ±1% 0.125W 2012 (0805 INCH)	R66, R67	CMP-066- 000175-2	Up to date	2	Yageo Group	RC0805FR-07100RL	Volume Production	RS	6185135	0.00454	0.00908
	0R	RES 0ohm ±1% 0.125W 2012 (0805 INCH)	R68, R70	CMP-066- 000150-2	Up to date	2	Vishay	CRCW08050000Z0EA	Volume Production	Newark	52K9721	0.022	0.044
	100k	RES 100kohm ±1% 0.125W 2012 (0805 INCH)	R69, R71	CMP-066- 000211-2	Up to date	2	Yageo Group	RC0805FR-07100KL	Volume Production	Newark	05M4536	0.014	0.14
	5011	Test Point, Black, Through Hole, RoHS, Bulk	TP1, TP2			2	Keystone	5011	Volume Production	Newark	88H2535	0.137	0.274
	5010	Test Point, Red, Through Hole, RoHS, Bulk	TP3, TP4			2	Keystone	5010	Volume Production	Mouser	534-5010	0.37	0.74
	ADG732BSUZ-REEL	1.8 to 5.5 V / +/- 2.5 V, 4 Ohm, 32- Channel Analog Multiplexer, -40 to 85 degC, 48-pin QFP (SU-48), Reel	U1			1	Analog Devices	ADG732BSUZ-REEL	Volume Production	Mouser	584-ADG732BSUZ-R	15.92	15.92
	74HC595PW,118	IC SHIFT REGISTER 8BIT 16TSSOP	U2			1	Nexperia	74HC595PW,118	Volume Production	Newark	75X1084	0.041	0.041
	2.54	PIN HEADER 2.54 1x10	X1	CMP-055- 000013-1	Up to date	1	Würth Elektronik	61300411121	Unknown	DigiKey	732-5317-ND	0.17494	0.17494

BOM Cost		PROJET:	2510_RegulationChauffage_Daugtherboard.PrjPcb										ETML-ES		Rue de Sébeillon 12 1004 Lausanne			
Date:	29.08.2025	Variante:	Quantité de production (nb PCBs) :										1	Quantité de production (nb PCBs) :		100		
Par:	LMS	Source:	2510_RegulationChauffage_Daugtherboard.BomDoc										Devi CHF		Devises :		CHF	
Line #	Designator	Value	Name	Manufacturer 1	Manufacturer Part Number 1	Quantity	Supplier 1	Supplier Part Number 1	Supplier Order Qty 1	Supplier Unit Price 1	Supplier Subtotal 1	Supplier Order Qty 1	Supplier Unit Price 1	Supplier Subtotal 1	Supplier Subtotal 1			
	C1, C2	0.1 uF	0.1 uF	Yageo Group	CC0805KRX7R9BB104	2	Farnell	3019949	2	0.0241	0.05	200	0.0241	0.05	4.82			
	D1, D2		SP1002-02JTG	Littelfuse	SP1002-02JTG	2	DigiKey	F4238CT-ND	2	0.1958	0.39	200	0.1958	0.39	39.17			
	D3, D4		SS14	onsemi	SS14	2	Farnell	1467537	2	0.2134	0.43	200	0.2134	0.43	42.68			
	J1		SFH11-PBPC-D10-RA-BK	Sullins	SFH11-PBPC-D10-RA-BK	1	Digkey	S9205-ND	1	0.9347	0.93	100	0.9347	0.93	93.47			
	J2, J3		5019	Keystone	5019	2	DigiKey	36-5019CT-ND	2	0.2231	0.45	200	0.2231	0.45	44.62			
	K1, K2		TQ2-5V-3	Panasonic	TQ2-5V-3	2	DigiKey	255-2257-ND	2	2.4900	4.98	200	2.4900	4.98	498.00			
	PCB1		PCB	Eurocircuit	-	1	Eurocircuit	-	1	55.6700	55.67	100	55.67	55.67	560.00			
	Q1, Q2		IRLML2402	UMWIRLML2402TR	UMWIRLML2402TR	2	Digikey	4518-UMWIRLML2402TR-ND	2	0.3480	0.70	200	0.3480	0.70	69.61			
	R1		200R	Bourns	3386H-1-2011LF	1	Farnell	9354930	1	1.2300	1.23	100	1.2300	1.23	123.00			
	R2		820R	Yageo Group	RC0805FR-13820RL	1	DigiKey	13-RC0805FR-13820RLCT-ND	1	0.0241	0.02	100	0.0241	0.02	2.41			
	R3, R4, R6, R8, R11, R15		6R2	Yageo Group	RC0805FR-076R2L	6	Farnell	3952295	6	0.0294	0.18	600	0.0294	0.18	17.64			
	R5, R7, R9, R10, R12, R13, R14, R16, R17, R18, R19, R20, R21, R22		6R8	Yageo Group	RC0805FR-076R8L	21	DigiKey	311-6.80CRCT-ND	21	0.0114	0.24	2100	0.0114	0.24	23.94			
	R24, R28, R31, R33		7R5	Yageo Group	AC0805FR-077R5L	4	DigiKey	13-AC0805FR-077R5LCT-ND	4	0.0196	0.08	400	0.0196	0.08	7.83			
	R66, R67	100	100R	Yageo Group	RC0805FR-07100RL	2	Farnell	9237364	2	0.0133	0.03	200	0.0133	0.03	2.66			
	R68, R70	0	0R	Vishay	CRCW08050000Z0EA	2	DigiKey	541-0.0ACT-ND	2	0.0178	0.04	200	0.0178	0.04	3.56			
	R69, R71	100k	100k	Yageo Group	RC0805FR-07100KL	2	Farnell	9237879	2	0.0125	0.02	200	0.0125	0.02	2.49			
	TP1, TP2		5011	Keystone	5011	2	Farnell	2431148	2	0.1159	0.23	200	0.1159	0.23	23.18			
	TP3, TP4		5010	Keystone	5010	2	DigiKey	36-5010-ND	2	0.2352	0.47	200	0.2352	0.47	47.03			
	U1		ADG732BSUZ-REEL	Analog Devices	ADG732BSUZ-REEL	1	Farnell	4032557	1	13.8800	13.88	100	13.88	13.88	680.00			
	U2		74HC595PW,118	Nexperia	74HC595PW,118	1	DigiKey	1727-3068-1-ND	1	0.2247	0.22	100	0.2247	0.22	22.47			
	X1	2.54	2.54	Würth Elektronik	61300411121	1	DigiKey	732-5317-ND	1	0.1404	0.14	100	0.1404	0.14	14.04			
											CHF 80.38			CHF 2 322.62				
Approved Notes																		